

Software Engineering Department

Braude College of Engineering

Final Project – Phase A (61998)

Sequential Eye-Tracking Analysis System for Dynamic Behavioral Pattern Identification in ASD

CODE: 26-2-D-1

By:

Lior Stein

Guy Shargorodsky

Advisor:

Dr. Anat Dahan

Dr. Julia Sheidin

Link to GitHub:
https://github.com/Lior-Stein/Sequential-Eye-Tracking-Analysis-System.git

Table of Contents

Abstract **3**
1 Introduction **3**
2 Background and Related Work **4**
2.1 Autism Spectrum Disorder (ASD) 4
2.2 Integration of ASD and Eye Tracking 5
2.3 Algorithmic Approaches for Dynamic AOI Classification 5
2.4 Automated AOI Extraction with VLM-GIST 6
3 Engineering Process **7**
3.1 Project Objectives 7
3.2 Development Process 8
3.2.1 Experimental Setup and Data Collection 8
3.2.2 System Implementation: The VLM-GIST Pipeline 9
3.2.3 System Evaluation and Behavioral Integration 11
3.2.4 Development Environment 11
3.3 System Requirements 12
3.3.1 Functional Requirements (FR) 12
3.3.2 Non-Functional Requirements (NFR) 13
4 Evaluation **14**
4.1 Evaluation Against the Manual Tobii Pro Lab Workflow 14
4.2 Expected Challenges & Limitations 15
5 GenAI Usage Discloser **15**
References: **15**

Abstract

Autism Spectrum Disorder (ASD) is a complex neurodevelopmental condition where early diagnosis is critical for improving long-term outcomes. While diagnosis heavily relies on subjective clinical observation, eye-tracking (ET) technology provides an objective alternative by capturing quantifiable data on visual attention. However, traditional ET solutions rely primarily on static metrics, such as fixation durations on stationary stimuli. These traditional approaches fail to capture the continuous, naturalistic gaze dynamics necessary to understand the underlying organization and strategy of a child's visual attention during complex interactions.

To address this limitation, this document presents an automated, end-to-end software system for sequential eye-tracking analysis. The proposed clinical support tool goes beyond static metrics by using neural networks to analyze dynamic gaze behaviors and continuous transitions between Areas of Interest (AOIs). Once this automated extraction is achieved, the next goal is to merge point-of-view gaze recordings with external room-camera footage capturing the child's physical play. Synchronizing these data streams will enable comprehensive behavioral analysis, specifically to evaluate whether a child visually engages with the toys they are physically manipulating. Ultimately, this research presents a practical, data-driven approach to analyzing sequential gaze behavior, contributing to the creation of accessible technological tools that support objective ASD diagnosis and intervention planning.

1 Introduction

Autism Spectrum Disorder (ASD) is a complex neurodevelopmental condition characterized by persistent challenges in social communication, social interaction, and restricted, repetitive patterns of behavior [Hirota et al., 2023]. The condition typically manifests in early childhood [Hodis et al., 2025], and as researchers emphasize, early diagnosis and personalized interventions are critical for improving long-term developmental outcomes [Hirota et al., 2023].

While diagnosis often relies on clinical observation of behavior, there is a growing need for objective, quantifiable biomarkers to support the diagnostic process. Eye-tracking (ET) technology has emerged as a particularly promising solution. It provides a non-invasive means of recording and analyzing eye movements, offering objective data on visual attention and cognitive processing patterns that underlie social understanding [Hessels et al., 2025].

The integration of ET into ASD research has confirmed its value. Studies consistently demonstrate that individuals with ASD display atypical visual attention patterns. For instance, children with ASD show reduced attention to social stimuli, such as faces, compared to their typically developing peers and often display a "circumscribed" or narrowly focused attention to specific non-social elements within a complex scene [Sasson et al., 2008]. These findings have been further validated by meta-analyses showing that gaze metrics can reliably distinguish between individuals with ASD and those without [Setien-Ramos et al., 2023].

However, despite these clear findings, translating ET from a research tool into a widely used clinical diagnostic instrument faces significant hurdles. As noted in recent comprehensive reviews, the field suffers from methodological challenges, including high variability in experimental protocols and a lack of standardized gaze metrics [Nguyen et al., 2025].

Furthermore, while advanced machine learning and deep learning algorithmsɲ such as SVM [Jaradat et al., 2025; Kaloforidis et al., 2025], LSTM networks [Ahmed et al., 2023; Al-Adhaileh et al., 2025], and CNNs [Ahmed et al., 2022], have shown high accuracy in classifying gaze data, a major limitation remains in the data extraction phase: the dynamic mapping of gaze coordinates to moving AOIs. Traditional workflows rely heavily on static snapshot definitions [Tobii Academy, n.d.] or manual frame-by-frame annotations, which are highly subjective, labor-intensive, and unable to scale for naturalistic, continuous play sessions [Jayawardena & Jayarathna, 2021; Tzamaras et al., 2023]. To bridge this gap and create unified, accessible clinical frameworks, these classification models must be preceded by automated pipelines that can process continuous video streams and dynamically map gaze coordinates without relying on restrictive static snapshots.

Therefore, the primary objective of this development project is to build such an automated pipeline. Specifically, it aims to automate AOI extraction using deep neural networks. Once these areas are successfully identified, the system will merge the point-of-view videos from the Tobii eye-tracker (which capture the child's gaze) with external room camera recordings (which capture the child's physical actions). Synchronizing these two video sources will allow us to understand a crucial behavioral link: whether the children are actually looking at the toys they are physically interacting with.

2 Background and Related Work

2.1 Autism Spectrum Disorder (ASD)

ASD typically emerges in early childhood and presents with a wide range of characteristics that vary in intensity and manifestation across individuals [Hodis et al., 2025]. Symptoms may include difficulties interpreting social cues, challenges with reciprocal communication, and repetitive behaviors or strong adherence to routines [Hodis et al., 2025]. Hirota et al. (2023) highlight the importance of early identification and tailoring interventions to the specific needs of each child, noting that timely support can significantly improve long-term developmental outcomes [Hirota et al., 2023].

ASD is recognized as a lifelong condition with diverse levels of required support. As emphasized by the Lancet Commission, improving diagnosis, care, and social inclusion requires coordinated, multidisciplinary efforts across the lifespan [Lord et al., 2021]. Epidemiological studies estimate a prevalence of approximately 1-2% globally, with observed increases largely attributable to heightened awareness and improved diagnostic practices [Hodis et al., 2025]. Recent advances in neuroscience and clinical research have also expanded the range of diagnostic and therapeutic tools available, including behavioral, digital, and emerging technological interventions aimed at both core and associated symptoms [Qin et al., 2024]. While no cure currently exists, continuous progress in early detection and evidence-based treatment continues to improve quality of life and functional outcomes for individuals on the spectrum.

2.2 Integration of ASD and Eye Tracking

The integration of ET technology into ASD research provides a powerful, objective biomarker that serves as a crucial alternative to lengthy, subjective clinical observations [de Belen et al., 2024]. To effectively translate this technology from a research instrument into a widespread clinical diagnostic tool, recent efforts focus on integrating eye tracking with advanced hardware and automated software pipelines. On the hardware front, integration involves moving away from restrictive, static monitor-based setups to capture continuous gaze dynamics in naturalistic environments [Chong et al., 2017; Noris et al., 2012]. Furthermore, to assess true face-to-face interactions and engagement with physical toys, integration now frequently involves wearable Point-of-View (POV) cameras and head-mounted eye trackers [Chong et al., 2017; Perkovich et al., 2024]. These head-mounted devices successfully tap into internal processes, capturing millisecond-level fluctuations in visual engagement without restricting a child's natural movements during play sessions [Perkovich et al., 2024]. Algorithmically, the raw sequential data collected from these devices is increasingly integrated into machine learning (ML) and deep learning (DL) frameworks to automate diagnosis and predict symptom severity [de Belen et al., 2024]. Rather than relying on simple static metrics like total fixation time, modern systems integrate Convolutional Neural Networks (CNNs), LSTM networks, and dynamic saliency models to process complex spatiotemporal gaze features and sequential transitions [Chong et al., 2017; de Belen et al., 2024]. This data-driven integration enables clinical support tools to interpret the continuous, dynamic environments critical for accurate ASD classification [de Belen et al., 2024].

The primary goal of integrating these advanced methodologies is to accurately quantify the atypical visual patterns characteristic of ASD. Individuals with ASD frequently display circumscribed attention to non-social elements and a "geometric preference," spending a significantly higher percentage of their time fixating on abstract, dynamic geometric images rather than social scenes [de Belen et al., 2024; Vargas-Cuentas et al., 2017]. They may also exhibit "sticky gaze"—a rigid fixation on a specific object accompanied by difficulty physically shifting the gaze to new targets, particularly social cues. Furthermore, semi-structured clinical tests using head-mounted trackers reveal that when interacting with physical toys (such as cars, balls, or wind-up toys), children with ASD not only look significantly less at an adult's face, but they also employ an eccentric viewing strategy [Noris et al., 2012]. When visually exploring these toys, children with ASD tend to direct their gaze lower and make much greater use of their lateral field of view than typically developing children [Noris et al., 2012].

Recent large-scale clinical trials have demonstrated that automated eye-tracking systems measuring these dynamic engagements can achieve diagnostic accuracy comparable to that of expert clinicians [Jones et al., 2023]. Building on these ideas, this project aims to create a fully automatic system to analyze how children shift their gaze and sustain long fixations. This will improve clinical evaluations for ASD.

2.3 Algorithmic Approaches for Dynamic AOI Classification

While wearable trackers provide rich and authentic data, they introduce significant analytical challenges. The participant is constantly moving their head and body; therefore, the

camera's axis and field of view shift continuously, resulting in a highly dynamic visual scene [Tzamaras et al., 2023]. In these unstructured environments, target objects and people frequently change position, scale, and orientation.

To address the complexities of these dynamic scenes without predefined tasks, researchers are increasingly adopting algorithmic approaches powered by Computer Vision and Machine Learning (CVML) [Tzamaras et al., 2023]. Deep learning architectures, such as Convolutional Neural Networks (CNNs), are employed to automatically detect and track moving AOIs [Chong et al., 2017; Jayawardena & Jayarathna, 2021]. These algorithms utilize pre-trained object detectors and instance segmentation models to dynamically generate bounding boxes or precise pixel-level masks for objects across video sequences [Jayawardena & Jayarathna, 2021]. Once the dynamic AOIs are algorithmically identified, the system can autonomously filter and map the raw spatiotemporal gaze coordinates onto these moving targets [Jayawardena & Jayarathna, 2021; Tzamaras et al., 2023]. This transition to algorithmic extraction is crucial for analyzing unstructured point-of-view recordings, as it successfully replaces the laborious manual annotation process and enables the objective evaluation of continuous gaze transitions in real-world settings. Ultimately, by providing a scalable and automated means of quantifying these complex visual patterns, these advanced frameworks lay the essential groundwork for using wearable eye-tracking as a reliable, data-driven tool for the early detection and classification of ASD [Ahmed et al., 2022; de Belen et al., 2024].

2.4 Automated AOI Extraction with VLM-GIST

To overcome the limitations of manual annotation, recent advancements in computer vision offer fully automated, zero-shot alternatives [Pätzold et al., 2025]. A prime example is the VLM-GIST (VLM Grounding for Instance Segmentation & Tracking) framework, which completely removes the need for manual prompting or prior scene knowledge [Pätzold et al., 2025]. To maintain real-time performance with minimal computational overhead when processing complex visual data, the execution of VLM-GIST is divided into two main components, as shown in Figure 1. The first is a low-frequency update mechanism that autonomously identifies objects and is lightweight, and the second is a high-frequency tracker that tracks them over time [Pätzold et al., 2025].

The update mechanism begins by utilizing general-purpose Vision-Language Models (VLMs) to analyze a single video frame and generate a "structured description" [Pätzold et al., 2025]. The VLM creates a machine-readable list, typically in JSON format, that catalogs every unique object visible in the scene [Pätzold et al., 2025]. Each entry includes essential attributes- specifically, an object name and a detailed natural-language description of its visual appearance, alongside any custom user-defined attributes [Pätzold et al., 2025]. Once generated, these rich descriptions serve as text prompts for an open-vocabulary object detector, which locates the corresponding objects within the image and draws bounding boxes around them to establish their spatial position [Pätzold et al., 2025]. To increase confidence, an optional validation step can also be applied at this stage; the system crops the bounding box regions and prompts the VLM again to confirm that the cropped image perfectly matches the original object description [Pätzold et al., 2025].

In the final step, these confirmed spatial detections are passed to a video segmentation model, such as SAM 2 [Pätzold et al., 2025]. The segmentation model translates the rectangular bounding boxes into precise, pixel-level masks [Pätzold et al., 2025]. Once initialization is complete, the lightweight tracking model continuously tracks these object masks in real time across the incoming video stream [Pätzold et al., 2025]. Because the tracker is highly efficient, it only requires the system to re-execute the more computationally intensive description and detection steps on demand when new objects enter the scene, making it a robust and scalable solution for continuous gaze analysis [Pätzold et al., 2025].

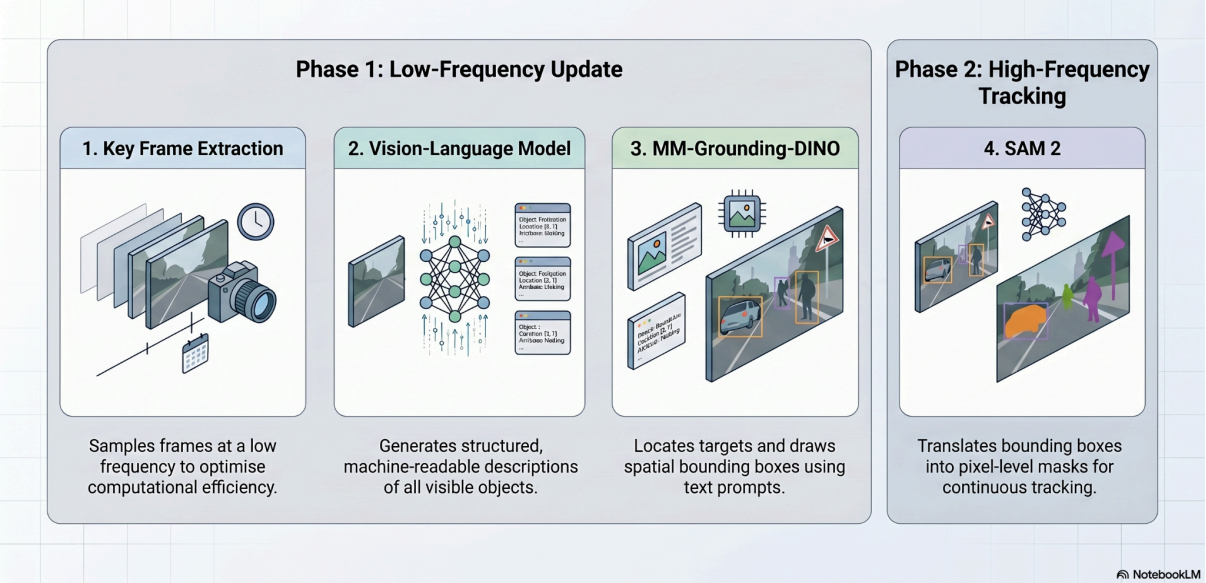


Figure 1: The VLM-GIST Pipeline Architecture

3 Engineering Process

3.1 Project Objectives

Current eye-tracking research increasingly utilizes wearable, head-mounted devices to capture authentic visual behaviors in naturalistic, unstructured environments [Noris et al., 2012; Tzamaras et al., 2023]. However, analyzing data from these dynamic scenes faces significant methodological challenges. Traditional analysis relies heavily on manually defining Areas of Interest (AOIs) using commercial software such as Tobii Pro Lab [Tobii Academy, n.d.]. Because the camera's axis and field of view shift continuously, manually annotating these dynamic AOIs frame-by-frame is tedious, highly subjective, and practically infeasible for large-scale use [Jayawardena & Jayarathna, 2021; Tzamaras et al., 2023].

To overcome these limitations, the primary objective of this project is to develop and implement an automated, end-to-end software system for dynamic AOI extraction and sequential eye-tracking analysis. Instead of relying on manual human ratings and static snapshots, this project will leverage advanced deep neural networks to automate data

extraction. Specifically, it uses the VLM-GIST (VLM Grounding for Instance Segmentation & Tracking) framework to autonomously identify, bound, and continuously track dynamic objects in real time without prior scene knowledge or manual prompting [Pätzold et al., 2025].

Once the dynamic AOIs are algorithmically identified, the system autonomously filters and maps the raw spatiotemporal gaze coordinates onto these continuously moving targets. Once the dynamic AOIs are algorithmically identified, the system autonomously filters and maps the raw spatiotemporal gaze coordinates onto these continuously moving targets. Initially, a core objective was to validate this approach by comparing the generated AOIs against a manual baseline mapped via Tobii Pro Lab methodologies. However, following discussions with our advisors, we concluded that a meaningful comparison would require developing a tool that directly integrates with Tobii's proprietary software, which is not practically feasible within the current scope of the project. Furthermore, we realized that a much more valuable next step for the researchers would be understanding the behavioral connection between the child's visual focus and their physical actions. Therefore, our subsequent objective shifted: the system will aim to merge the gaze data with external room footage to determine whether the child is actually looking at the toy they are actively playing with.

3.2 Development Process

The development process for this automated eye-tracking analysis system follows a modular approach, structured into three sequential phases.

First, the experimental setup uses a dataset of dynamic video recordings, collected in collaboration with researchers at the University of Haifa, that combines head-mounted gaze tracking with external room cameras to capture naturalistic behavioral patterns. Second, the system implementation details the construction of the VLM-GIST pipeline, which is designed to fully automate dynamic AOI extraction and gaze mapping. Finally, the system focuses on behavioral integration by synchronizing these multi-modal streams to accurately link the child's visual attention with their physical actions and movements in the room.

3.2.1 Experimental Setup and Data Collection

This research, conducted in collaboration with researchers from the University of Haifa, focuses on identifying differences in visual attention patterns between children with ASD and those without.
The experiment will take place in a dedicated playroom environment. In each experimental session, a single child, either diagnosed with ASD or typically developing, will enter the room individually. Each child will have 10 minutes of free playtime in the room. The playroom will be divided into three main spatial regions, each defined as an Area of Interest (AOI). Within each AOI, multiple sub-AOIs will be defined to represent specific objects or functional elements within the play area. Figures 2-4 illustrate the experimental stimuli included in these main regions: games placed on a table (Figure 2), a rubber tube together with a slide and a wall-mounted climbing rack (Figure 3), and a slinky placed on a cushion (Figure 4). The definition and labeling of AOIs and sub-AOIs will be determined in advance by the researchers from the University of Haifa, based on clinical relevance and experimental design considerations.

During the experiment, the child will wear Tobii eye-tracking glasses, which will continuously record eye movements while the child engages freely with the play environment. The setup allows for naturalistic behavior without direct instructions, enabling the capture of spontaneous visual exploration patterns.
The study was approved, as required, by the Institutional Research Ethics Committee at the University of Haifa. The participants provided informed consent to participate in this study.



Figure 2: Games placed on a table.

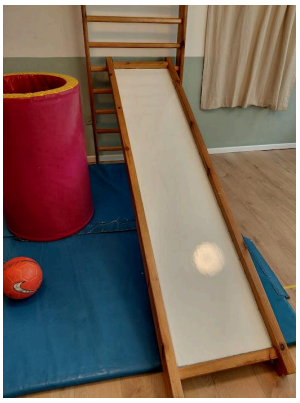


Figure 3: Rubber tube and a slide connected to a climbing rack.



Figure 4: A slinky placed on a cushion.

3.2.2 System Implementation: The VLM-GIST Pipeline

The core of the automated system relies on the implementation of the VLM-GIST framework, which processes unstructured visual data into quantifiable Areas of Interest (AOIs), as illustrated in the 6-step logical flowchart in Figure 5. Because continuously querying deep neural networks on every single video frame is computationally prohibitive, the pipeline architecture is divided into a low-frequency update mechanism and a high-frequency tracking module [Pätzold et al., 2025]. The process begins with a data ingestion module that takes raw egocentric video streams and continuous spatiotemporal gaze coordinates as inputs. After synchronizing the video recordings with the eye-tracking data, the system optimizes computational efficiency by operating on "Key Frames"- specific, isolated images sampled from the video stream at low frequencies, such as at regular intervals or when new object instances enter the field of view [Pätzold et al., 2025].

These isolated Key Frames serve as the primary input for the Vision-Language Model (VLM). The VLM scans the key frame to autonomously generate a machine-readable structured description of the scene without requiring any prior knowledge [Pätzold et al., 2025]. To achieve this, the model is strictly prompted to output a JSON-formatted list of dictionaries representing all visible objects. Within this JSON structure, each dictionary entry must include two attributes: object_name and description. The description is deliberately limited to 10 words that detail the object's type, color, and visual appearance [Pätzold et al., 2025]. Furthermore, the JSON can include custom boolean attributes (e.g., is_relevant_to_task: true/false) generated via decoupled attribution to filter out background noise [Pätzold et al., 2025]. The output of this stage is a structured JSON object that contains precise semantic descriptions of every relevant target in the frame.

Once the JSON is generated, the pipeline moves to the detection and segmentation phase. The key frame image and the text-based description attributes extracted from the JSON are fed as inputs into an open-vocabulary object detector (such as MM-Grounding-DINO) [Pätzold et al., 2025]. The detector uses these text prompts to locate the described items within the image, outputting 2D spatial bounding boxes around each identified object. To handle the continuous movement of the child's head and objects during free play, these bounding boxes, along with the continuous high-frequency video stream, are fed into a video segmentation model, specifically the Segment Anything Model 2 (SAM 2) [Pätzold et al., 2025]. SAM 2 translates the rectangular boxes into precise, pixel-level masks and outputs continuously tracked, dynamic masks with unique Track IDs across all subsequent high-frequency video frames.

Following the VLM-GIST tracking output, the final stage of our system involves computational gaze mapping. This module takes the dynamically tracked pixel-level masks (the moving AOIs) and the synchronized raw gaze coordinates as inputs. A computational mapping algorithm programmatically intersects the gaze coordinates with the tracked masks, frame by frame. The ultimate output of this integrated pipeline is a set of sequential behavioral metrics that autonomously determine exactly which dynamic AOI is being fixated on at each millisecond, thereby eliminating the need for manual frame-by-frame annotation.

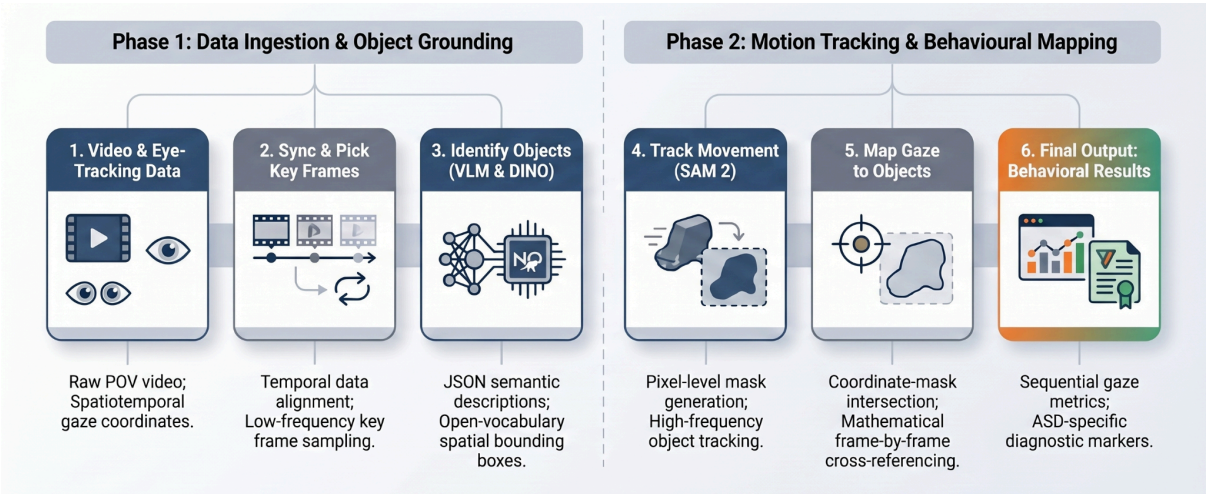


Figure 5: Logical Process of the VLM-GIST Eye-Tracking Integration

3.2.3 System Evaluation and Behavioral Integration

The final stage of the project will involve a comprehensive evaluation phase to assess the system's ability to extract dynamic AOIs and synchronize multi-modal video streams. The evaluation will focus on the system's practical utility for behavioral research. This analysis will target two key dimensions: tracking robustness and synchronization accuracy. Tracking robustness will be evaluated by the system's ability to maintain stable, continuous object masks during dynamic, naturalistic play sessions without excessive computational delays. Synchronization accuracy will be tested by verifying that the temporal gaze coordinates from the eye-tracking glasses correctly align with the physical actions captured by the external room camera. Ultimately, by demonstrating that the automated pipeline can reliably link a child's visual attention to their physical engagement, this phase will prove the system's viability as an objective, scalable tool for analyzing complex behavioral patterns in ASD.

3.2.4 Development Environment

The development and implementation of the automated eye-tracking pipeline will use a modern computer vision technology stack centered on the VLM-GIST framework, as shown in Figure 6. The core infrastructure of the system will be developed using Python and structured around the Robot Operating System, specifically ROS2 (Jazzy), which serves as the central middleware for managing interactions among the models. To handle the computational load and GPU acceleration, the architecture relies on dedicated inference servers. Local execution of the Vision-Language Models is orchestrated using vLLM servers, while the open-vocabulary object detector (MM-Grounding-DINO) and the intensive video segmentation network (SAM 2) for dynamic, pixel-level tracking are hosted via NimbRo Vision Servers.

For data ingestion, manipulation, and structural formatting, the pipeline will heavily utilize the FiftyOne computer vision toolset. The system will use specialized ROS2 nodes to interact with FiftyOne datasets to efficiently download, describe, detect, and validate visual streams and dynamic tracking masks. Day-to-day pipeline execution and the orchestration of these processing stages will be handled primarily through Jupyter Notebooks and Python scripts that interface directly with ROS2 nodes.

The software engineering process will be managed using an Integrated Development Environment (IDE) such as Visual Studio Code (VS Code), which facilitates modular architecture design and the native execution of the system's Python-based Jupyter Notebooks. To ensure structural reliability and resolve complex system dependencies without conflicts, development will be executed using VS Code's DevContainers, which containerize the entire ROS2 deployment environment using Docker. Source code version control will be strictly maintained via Git, with the central repository hosted on GitHub.
Finally, to support the subsequent goal of behavioral analysis, the pipeline will incorporate data synchronization scripts to accurately merge temporal gaze data from Tobii point-of-view recordings with spatial video data from external room cameras. This integration will enable the system to automatically evaluate the connection between the child's visual focus and their physical interactions.

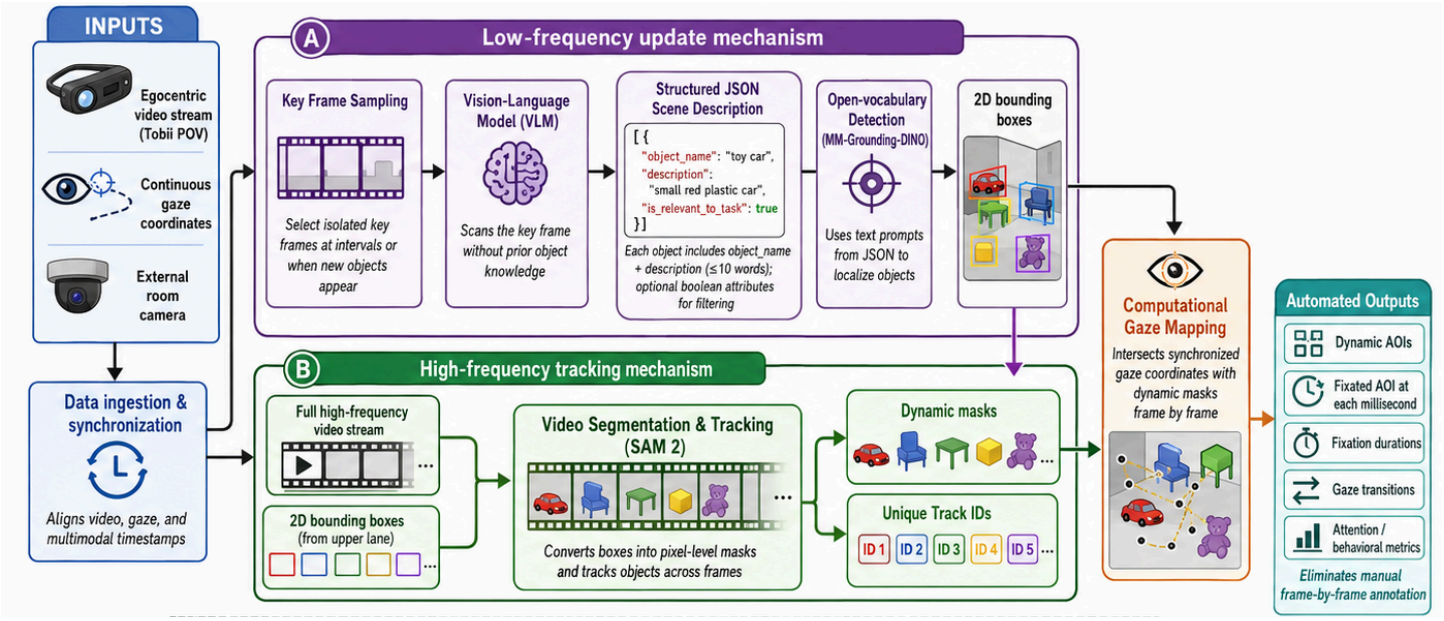


Figure 6: System Architecture of the Automated Eye-Tracking Pipeline

3.3 System Requirements

Based on the system's goals, the system design is detailed in terms of functional processing capabilities and evaluated against standardized non-functional quality attributes.

3.3.1 Functional Requirements (FR)

The functional requirements outline the fundamental system operations, data transformations, and analytical workflows necessary to deliver an automated clinical support tool.

- **FR-1: Data Ingestion:** The system must import video streams and raw gaze coordinate data.
- **FR-2: Data Synchronization:** The system must synchronize video recordings with gaze coordinate data.
- **FR-3: Key Frame Extraction:** The system must extract key frames from the video stream.
- **FR-4: Object Identification:** The system must identify and describe visible objects within a frame.
- **FR-5: Data Formatting:** The system must format the object descriptions into a structured list.

- **FR-6: Object Bounding:** The system must spatially bound the identified objects.
- **FR-7: Object Tracking:** The system must track bounded objects across consecutive video frames.
- **FR-8: Gaze Mapping:** The system must map gaze coordinates.
- **FR-9: Metrics Calculation:** The system must calculate gaze metrics.
- **FR-10: Baseline Comparison:** The system must compare system outputs.
- **FR-11: Data Export:** The system must export the calculated metrics.

3.3.2 Non-Functional Requirements (NFR)

These requirements define the quality attributes and performance characteristics of the system, based on the project's scope and challenges:

- **NFR-1: Performance:** The system must process video streams and track dynamic AOIs in real time.
- **NFR-2: Scalability:** The automated pipeline must efficiently analyze continuous, multi-frame gaze data across large participant pools.
- **NFR-3: Usability:** The system shall support inputting custom text attributes or target-specific objects without needing prior scene knowledge, manual bounding box drawing, or retraining of the underlying machine learning models.
- **NFR-4: Reliability:** The integrated neural networks must reliably quantify atypical visual attention patterns—such as circumscribed attention, sticky gaze, and eccentric viewing strategies.
- **NFR-5: Ingestion Compatibility:** The system must support standard raw data formats generated by Tobii commercial head-mounted eye trackers.
- **NFR-6: Data Security and Privacy:** The system will adhere to all relevant data privacy and protection regulations for handling sensitive child behavioral and potential medical data.
- **NFR-7: Execution Triggers:** Key frame sampling must be executed at a defined interval or in response to a specific update trigger.
- **NFR-8: Detector Architecture:** The system must strictly utilize an open-vocabulary object detector.

- **NFR-9: Segmentation Architecture:** The system must strictly utilize a video segmentation network.

- **NFR-10: Temporal Processing Resolution:** The system must continuously track dynamic AOIs in real time and process raw gaze coordinates at the millisecond level.

- **NFR-11: Segmentation Precision:** The system must translate bounding boxes into precise, pixel-level tracking masks.

- **NFR-12: Mapping Methodology:** The system must map the gaze coordinates onto the tracked masks using strict mathematical cross-referencing.

- **NFR-13: Downstream Data Compatibility:** The system must export the calculated dynamic gaze metrics in a structured format specifically designed for subsequent behavioral pattern classification models.

- **NFR-14: Comparative Benchmarking Rigor:** The system's automated outputs must be systematically contrasted against static snapshot baselines (e.g., Tobii Pro Lab manual annotations) to formally validate computational efficiency and structural reliability.

- **NFR-15: Data Format:** The system must generate a structured list of descriptions in machine-readable JSON format.

4 Evaluation

4.1 Evaluation Against the Manual Tobii Pro Lab Workflow

Tobii Pro Lab is a traditional eye-tracking software platform used for recording gaze data and manually defining Areas of Interest (AOIs). While this approach can work well for static images, it becomes time-consuming and subjective when applied to dynamic videos, especially when objects and the camera view change continuously.

In this project, Tobii-based manual annotation will serve as a reference point for evaluating the proposed automated system. We will compare the time required by our automated method to detect and track AOIs with the time required for manual Tobii-based annotation. This will help determine whether the system reduces manual work and improves processing efficiency. After the AOIs are detected, the system will match the child's gaze points to the corresponding AOIs over time. To evaluate this, we will compare the system's gaze-to-AOI matches with a manually checked sample. The main metric will be the percentage of gaze points correctly matched to AOIs.

By evaluating both processing time and gaze-to-AOI matching accuracy, we can assess whether the system is efficient, reliable, and useful for analyzing natural play behavior in ASD research.

4.2 Expected Challenges & Limitations

Developing this system comes with several technical challenges. The biggest challenge is syncing the two different video sources: the Tobii eye-tracking glasses and the room camera. Because the cameras might start recording at different times and have different settings, we need to match them perfectly. This is crucial to ensure that the child's gaze accurately aligns with their physical actions.

Another challenge is the natural play environment. Fast movements, a shaking camera, or the child's hands hiding the toys can confuse the AI and make the system lose track of the objects.

Finally, a major limitation is the substantial computational power required. Running advanced AI models on continuous video streams is very resource-intensive. It requires robust hardware and significant processing time to work properly.

5 GenAI Usage Discloser

During the preparation of this work, the authors used Gemini and ChatGPT to enhance language and grammar. After using this service, the authors reviewed and edited the content as needed and took full responsibility for the project's content.

References:

1. **Ahmed, Z. A. T., Albalawi, E., Aldhyani, T. H. H., Jadhav, M. E., Janrao, P., Obeidat, M. R. M. (2023).** Applying Eye Tracking with Deep Learning Techniques for Early-Stage Detection of ASD. Information, 8(11), 168. https://doi.org/10.3390/data8110168

2. **Jaradat, A. S., Wedyan, M., Alomari, S., Barhoush, M. M. (2025).** Using Machine Learning to Diagnose Autism Based on Eye Tracking Technology. Diagnostics, 15(1), 66. https://doi.org/10.3390/diagnostics15010066

3. **Al-Adhaileh, M. H., Alsubari, S. N., Al-Nefaie, A. H., Ahmad, S., Alhamadi, A. A. (2025).** Diagnosing autism spectrum disorder based on eye tracking technology using deep learning models. Frontiers in Medicine, 12, 1690177. https://doi.org/10.3389/fmed.2025.1690177

4. **Antolí, A., Rodriguez-Lozano, F. J., Cañas, J. J., Vacas, J., Cuadrado, F., Sánchez-Raya, A., Pérez-Dueñas, C., Gámez-Granados, J. C. (2025).** Using explainable machine learning and eye-tracking for diagnosing autism spectrum and developmental language disorders in social attention tasks. Frontiers in Neuroscience, 19, 1558621. https://doi.org/10.3389/fnins.2025.1558621

5. **Ahmed, I. A., Senan, E. M., Rassem, T. H., Ali, M. A. H., Shatnawi, H. S. A., Alwazer, S. M., Alshahrani, M. (2022).** Eye Tracking-Based Diagnosis and Early Detection of Autism Spectrum Disorder Using Machine Learning and Deep Learning Techniques. Electronics, 11(4), 530. https://doi.org/10.3390/electronics11040530

6. **Nguyen, T. Q. H., Nguyen, T. K. C., Le, T. H., Ngo, T. D. (2025).** Eye-tracking technology applications for supporting individuals with autism spectrum disorder: insights, challenges, and opportunities. Disability and Rehabilitation: Assistive Technology, 20(8), 2630-2665. https://doi.org/10.1080/17483107.2025.2531236

7. **Hessels, R. S., Nuthmann, A., Nyström, M., Andersson, R., Niehorster, D. C., Hooge, I. T. C. (2025).** The fundamentals of eye tracking part 1: The link between theory and research question. Behavior Research Methods, 57, 16. https://doi.org/10.3758/s13428-024-02544-8

8. **Hirota, T., King, B. H. (2023).** Autism Spectrum Disorder: A Review. JAMA, 329(2), 157-168. https://doi.org/10.1001/jama.2022.23661

9. **Hodis, B., Mughal, S., Saadabadi, A. (2025).** Autism Spectrum Disorder. StatPearls Publishing. https://www.ncbi.nlm.nih.gov/books/NBK525976/

10. **Kaloforidis, N., Kollias, K.-F., Radoglou-Grammatikis, P., Sarigiannidis, P., Fragulis, G. F. (2025).** Autism Spectrum Disorder Classification in Children Using Eye-Tracking Data and Machine Learning. Engineering Proceedings, 107, 12. https://doi.org/10.3390/engproc2025107012

11. **Lord, C., et al. (2021).** The Lancet Commission on the future of care and clinical research in autism. The Lancet, 399(10321), 271-334. https://doi.org/10.1016/S0140-6736(21)01541-5

12. **Qin, L., Wang, H., Ning, W., Cui, M., Wang, Q. (2024).** New advances in the diagnosis and treatment of autism spectrum disorders. European Journal of Medical Research, 29, 322. https://doi.org/10.1186/s40001-024-01916-2

13. **Sasson, N. J., Turner-Brown, L., Holtzclaw, T., Lam, K. S. L., Bodfish, J. W. (2008).** Children with autism demonstrate circumscribed attention during passive viewing of complex social and nonsocial picture arrays. Autism Research, 1(1), 31–42. https://doi.org/10.1002/aur.4

14. **Setien-Ramos, I., Lugo-Marín, J., Gisbert-Gustemps, L., Díez-Villoria, E., Magán-Maganto, M., Canal-Bedia, R., Ramos-Quiroga, J. A. (2023).** Eye-Tracking Studies in Adults with Autism Spectrum Disorder: A Systematic Review and Meta-analysis. Journal of Autism and Developmental Disorders, 53(6), 2430-2443. https://doi.org/10.1007/s10803-022-05524-z

15. **Chong, E., Chanda, K., Ye, Z., Southerland, A., Ruiz, N., Jones, R. M., Rozga, A., Rehg, J. M. (2017).** Detecting Gaze Towards Eyes in Natural Social Interactions and Its Use in Child Assessment. PACM Interact. Mob. Wearable Ubiquitous Technol., 1(3), Article 43. https://doi.org/10.1145/3131902

16. **de Belen, R. A. J., Eapen, V., Bednarz, T., Sowmya, A. (2024).** Using visual attention estimation on videos for automated prediction of autism spectrum disorder and symptom severity in preschool children. PLoS ONE, 19(2): e0282818. https://doi.org/10.1371/journal.pone.0282818

17. **Jayawardena, G., Jayarathna, S. (2021).** Automated Filtering of Eye Movements Using Dynamic AOI in Multiple Granularity Levels. International Journal of Multimedia Data Engineering and Management, 12(1), 49-64. https://doi.org/10.4018/IJMDEM.2021010104

18. **Jones, W., Klaiman, C., Richardson, S., Aoki, C., Smith, C., Minjarez, M., Bernier, R., Pedapati, E., Bishop, S., Ence, W., Wainer, A., Moriuchi, J., Tay, S.-W., & Klin, A. (2023).** Eye-tracking–based measurement of social visual engagement compared with expert clinical diagnosis of autism. JAMA, 330(9), 854-865.
https://doi.org/10.1001/jama.2023.13295

19. **Noris, B., Nadel, J., Barker, M., Hadjikhani, N., Billard, A. (2012).** Investigating Gaze of Children with ASD in Naturalistic Settings. PLoS ONE, 7(9): 0044144.
https://doi.org/10.1371/journal.pone.0044144

20. **Perkovich, E., Laakman, A., Mire, S., Yoshida, H. (2024).** Conducting head-mounted eye-tracking research with young children with autism and children with increased likelihood of later autism diagnosis. Journal of Neurodevelopmental Disorders, 16:7.
https://doi.org/10.1186/s11689-024-09524-1

21. **Rigby, S. N., Stoesz, B. M., & Jakobson, L. S. (2016).** Gaze patterns during scene processing in typical adults and adults with autism spectrum disorders. Research in Autism Spectrum Disorders, 25, 24–36.
https://doi.org/10.1016/j.rasd.2016.01.012

22. **Tobii Academy. (n.d.).** Foundations of Eye Tracking: Complementary course e-book.

23. **Tzamaras, H. M., Wu, H.-L., Moore, J. Z., Miller, S. R. (2023).** Shifting Perspectives: A proposed framework for analyzing head-mounted eye-tracking data with dynamic areas of interest and dynamic scenes. Proc Hum Factors Ergon Soc Annu Meet.
https://doi.org/10.1177/21695067231192929

24. **Vargas-Cuentas, N. I., Roman-Gonzalez, A., Gilman, R. H., Barrientos, F., Ting, J., Hidalgo, D., Jensen, K., Zimic, M. (2017).** Developing an eye-tracking algorithm as a potential tool for early diagnosis of autism spectrum disorder in children. PLoS ONE, 12(11): e0188826.
https://doi.org/10.1371/journal.pone.0188826

25. **Pätzold, B., Nogga, J., & Behnke, S. (2025).** Leveraging Vision-Language Models for Open-Vocabulary Instance Segmentation and Tracking. IEEE Robotics and

Automation Letters.
https://doi.org/10.1109/LRA.2025.3606363